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E-101 GENERATOR

1. SCOPE OF WORK

The work shall include the design, supply, manufacturing, factory testing, delivery to the site, installation, commissioning and testing of Generator in accordance with Part 4, Subsection 1.2 Summary of Work.

Steam Turbine Generator (STG) shall be Three-phase, 50 Hz, 3000 r/min completed with excitation system. The generator cooling shall be either hydrogen or closed circuit air cooling system. The Bidder shall prove that the generator is capable to operate as a base load plant unit i.e minimum of 6000 hours per annum. The above shall be complete all accessories required for a functional system specified but not be limited to:

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- (1) Generator hydrogen cooling system
- (2) Hydrogen seal oil system
- (3) Generator closed air cooling system
- (4) Sole plates, subsole plates and final grouting

2. CODES AND STANDARDS

The Contractor shall comply with the requirements of PART 4, Subsection 2.5. The following codes and standards are specifically applicable to the design, manufacturing and testing of the equipment included in this Specification:

2.1 National

- PUIL : Peraturan Umum Instalasi Listrik - Indonesia (General Regulation of Electrical Installation Indonesia).
- SPLN : Standard Perusahaan Umum Listrik Negara.

2.2 International Electrotechnical Commission

- IEC 60034 : Rotating electrical machines
- IEC 60044 : Instrument transformers

IEC 60050-411	:	International electrotechnical vocabulary chapter 11 – Rotating machinery
IEC 60076	:	Power transformers
IEC 60137	:	Insulated bushings for alternating voltages above 1000V
IEC 60186	:	Voltage transformers
IEC 60296	:	Fluids for electrotechnical applications – Unused mineral insulating oils for transformer and switchgear
IEC 60529	:	Degrees of protection provided by enclosures (IP Code)

2.3 American National Standards Institute and Institute of Electrical and Electronics Engineers

ANSI/IEEE 95	:	Recommended Practice for Insulation Testing of Large AC Rotating Machinery with High Direct Voltage
ANSI/IEEE 115	:	Test Procedures for Synchronous Machines.
ANSI/IEEE 421.1:		Standard Definitions for Excitation Systems for Synchronous Machines
ANSI/IEEE 421.2:		Guide for Identification, Testing and Evaluation of Dynamic Performance of Excitation Control Systems
ANSI/IEEE C37.90.1	:	Guide for Surge Withstand Capability (SWC) Tests
ANSI/IEEE C 50.10	:	Rotating Electrical Machinery-Synchronous Machines

2.4 National Electrical Manufacturers Association

NEMA MG 1	:	Motors and Generators National Fire Protection Association
NFPA 70	:	National Electrical Code

2.5 Others

AFBMA	:	Anti Friction Bearing Manufacturer's Association.
ASME	:	American Society for Mechanical Engineers
ASTM	:	American Society for Testing and Materials

HEI	:	Heat Exchange Institute Standard for Power Plant Heat Exchangers
HIS	:	Hydraulic Institute Standards
OSHA	:	Occupational Safety and Health Administration
NERC	:	North American Electric Reliability Council

3 OPERATING AND DESIGN CONDITIONS

The generator shall be designed for indoor operation in non-air-conditioned hall in a tropical climate along the sea coast. Refer to Part 4, Subsections 1.1, 2.14 and 2.16 for meteorological, seismic and environmental conditions at the site.

The generator shall be properly designed for application directly coupled to steam turbine as its prime mover.

4 DETAILED REQUIREMENTS

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4.1 Generator and Appurtenances

The generator shall be supplied with all necessary auxiliary equipment to make a completely functional system and shall include, but not be limited to the followings:

- (1) Excitation system complete
- (2) Generator auxiliary systems
- (3) Turbine generator common lubricating oil system
- (4) Core monitor (For H2 cooled generator)
- (5) Generator stator housing
- (6) Bearing pedestals or brackets
- (7) Phase and neutral terminals and current transformers
- (8) Terminal enclosure
- (9) Rotor withdrawal and insertion gear, all special tools and instruments
- (10) All required local instrumentation and controls

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4.2 Generator Rating

The guaranteed continuous rating of the generator shall match the prime mover output at 0.80 power factor and at the maximum permissible air temperature or rated hydrogen (H₂) pressure (if required). The generator shall have the following additional ratings:

- (1) Rated power factor: 0.80 lagging
- (2) Voltage: Manufacturer's optimum
(preferably in 10-16 kV range)
- (3) Number of phases: 3
- (4) Frequency.....: 50 Hz
- (5) Speed: 3000 r/min
- (6) Short circuit ratio.....: Not less than 0.51
- (7) Negative sequence capability (I₂)²T: Minimum 10
- (8) Three -phase short circuit capability: according to IEC
- (9) Excitation system response ratio: 2.0
- (10) Nominal excitation voltage: Manufacturer's optimum
- (11) Stator winding connection: Star
- (12) Field winding conductors: Copper
- (13) Winding insulation class: Class F

4.3 Generator Characteristics

The generator shall be of the cylindrical rotor type. The generator shall be designed to operate successfully and to develop the capacities specified under the operating conditions described.

4.3.1 Electrical Characteristics

The generator shall be capable of operating continuously at the rated kVA load over the power factor range 0.80 lagging to 0.95 leading without loss of stability and control under all operational conditions defined in this specification. The wave form of the generated voltage, measured from line to line on open circuit at rated voltage and speed shall have a deviation factor of

not more than 10 % from a pure sine wave. Generator shall have a balanced Telephone Influence Factor (T.I.F.) of not more than 40, and a residual T.I.F. of not more than 30.

The generator shall be capable of operating continuously at maximum rated kVA output at any power factor between the rated 0.80 power factor lagging and 0.95 leading and at any voltage between five percent (5%) below and five percent (5%) above rated voltage, within guaranteed temperature limits.

The generator rotor shall be of the revolving field and non-salient pole type.

The generator capability diagram shall show power curves of the turbine and any restrictions imposed by the associated equipment such as excitation and governing systems, as well as inherent generator capability. Curves demonstrating the performance at increased or reduced air temperature or hydrogen pressure (if required), and with a reduced numbers of hydrogen coolers in service shall also be provided. With one cooler out of service the generator shall be capable of supplying at least 67 % of rated MVA output according to IEC without exceeding the specified generator temperatures. In this case, however, the cooling gas temperature may exceed the specified value for service with all coolers operational.

The generator shall capable of operating continuously at loads corresponding to the normal and maximum possible turbine power produced. The Contractor shall state the maximum allowable air temperatures at which the generator may operate without exceeding rated temperatures while delivering this maximum power and maintaining 0.80 power factor. The Contractor shall furnish generator capability curves at rated air temperature or rated hydrogen pressure (if required), and at the maximum allowable operating time under his condition without exceeding rated generator temperatures.

The generator shall be suitable in every respect for continuous and later cycling including two-shift operation on the PLN Network system.

The Contractor shall state in this Bid the values of the generator reactance as follows:

- (1) Maximum value of synchronous reactance

- (2) Maximum value of transient reactance
- (3) Maximum value of sub-transient reactance
- (4) The above values shall be quoted inclusive of all tolerances.

The design tolerance on the values of these reactances shall be not more than plus 10 % for the synchronous and not more than plus or minus 15 % for the transient and sub-transient reactances. The Contractor shall guarantee the generator reactances and their tolerances.

The generator shall be capable of continuous safe operation and with a negative sequence stator current of 10 % of rated stator current, and the following limits:

- (1) Rated power factor for positive sequence current component.
- (2) Stator positive sequence voltage at terminals not more than 100 % rated voltage.
- (3) Stator positive sequence current not more than 90 % rated current.
- (4) The stator and rotor winding temperatures shall not exceed the design temperature for rated output.

4.3.2 Insulation and Temperature Rises

The generator rotor and stator windings and connections including phase and neutral leads shall have Class F insulation. The temperature rises and total temperatures shall not exceed those permitted for Class B insulation.

The allowable temperature rises shall be reduced if the generator coolant cold temperature is above 40°C.

4.4 Generator Core Monitor

For H₂ cooled generator, A core monitor and pyrolyzate collector shall be provided to detect and identify the thermal decomposition of insulating materials in the cooling medium. The equipment shall be sensitive enough to detect insulation damage at an early stage.

4.5 Abnormal Operating Conditions

The turbine generator shall be capable of withstanding the following abnormal operating conditions without harmful deformation of the windings.

- (1) Faulty synchronization, such as out of phase synchronization, frequency and voltage mis-match.
- (2) Sudden short circuit at the generator terminals from rated load and 105 % voltage
- (3) High speed single and slow speed three pole reclosing of the high voltage switchgear following system faults.
- (4) Sub-synchronous resonance
- (5) The effects of system harmonic frequency currents caused by loads such as large arc furnaces located in close proximity to the plant in accordance with IEC codes.

4.6 Vibration

Vibration monitoring equipment, preferably with the same type as the generator prime mover shall be provided for each main shaft bearing of the generator set.

Vibration measurements shall also be made and recorded of the main bearings and other parts according to the IEC standard in the vertical, axial and horizontal directions.

The maximum permissible vibration of the stator core and windings shall be 25 micron peak to peak and no natural resonance shall occur between the frequency ranges of 40-60 HZ and 85 -115 HZ. The Contractor shall submit sufficient data or test results to demonstrate that this requirement will be met.

The Contractor shall also submit the vibration test report performed during the commissioning showing all vibration measurements including those when running through critical speeds and up to rated power.

4.7 Stator Housing

The stator housing shall be of heavy fabricated rolled steel plates, supported internally by axial and circumferential ribs.

The generator stator housing design shall ensure that pockets of gas (hydrogen if applicable) will not be formed in any location inside the housing during purging operations. The generator design shall allow convenient removal and replacement of the stator and/or the rotor.

The Contractor shall also furnish torque load data for the generator corresponding to sudden short circuit at the generator terminal or 180 out of phase synchronizing.

4.8 Stator Core

The generator stator core shall use steel stampings of high permeability and low hysteresis loss and the windings shall be firmly embedded in slots or tunnels in the core. The stator core and windings including end rings shall be completely factory assembled.

The stator core design shall minimize the effects of the inherent 100 Hz vibrations due to the action of the field system. The core shall be mounted such that these vibrations are not transmitted to bearings, hydrogen seals, or to the turbine generator foundations. A spring mounted core is preferred except where the Contractor can show from relevant test results that spring mounting is not necessary and that the proposed design is suitable for the foundations and will not affect turbine vibration.

The stator core and its attachments to the casing shall be designed to withstand the specified sudden short circuit and fault conditions.

The core laminations shall be cold rolled steel, and insulated with varnish. The use of paper based or other hygroscopic material whether between successive or batches of laminations is not acceptable.

A minimum of six (6) temperature detectors shall be furnished on the stator end iron laminations.

4.9 Stator Winding

Two (2) layer windings with single-turn coils are preferred.

A fully integrated insulation system that includes mica and epoxy or polyester based materials subjected to rigorous quality control standards is preferred to those using asphalt, shellac or nylon based materials for the stator winding ground insulation. The surfaces of the slot portion of the coils shall be coated with materials based on graphite, silicon carbide or the Contractor's equivalent to prevent corona. Stress grading shall also be provided on the coil sides where they emerge from the end of the core slots.

Coil to coil connections and ends shall be solidly brazed and fully insulated. All other live conductors or parts shall also be insulated. The use of hygroscopic insulation materials, which may absorb water from the hydrogen, air or fire protection medium, shall not be permitted.

In the end winding region the use of conducting materials sharp corners or projections shall be kept to a minimum. Any metallic fastener in this region shall be locked with approved locking devices.

The generator shall be braced to withstand the fault current which will result from any type of short circuit at the terminals provided the duration of the short circuit is not sufficient to cause injurious heating. The generator neutral shall be grounded through a neutral grounding transformer with a secondary resistor to limit the phase to ground fault current in the winding.

The Contractor shall include stator insulation life curves showing breakdown voltage from coil tests as a function of time to failure. The data shall be relevant to the coil size and shape, temperature and insulation type and treatment of the generator offered. The Bidder shall also include recommended "in service" winding dielectric tests (for example loss tangent or equivalent) on the wound machine.

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The insulation resistance, R_m , of the entire winding after the winding dryout, measured in mega ohms and referred to 40°C, shall not be less than the followings as per IEEE or equivalent.

$$R_m = kV + 1 \text{ (mega ohm)}$$

Where kV = generator line to line rated voltage

A suitable coil shall be installed in a stator slot near the middle of the core, or equivalent method, to determine the generator wave form by oscillograph.

4.10 Stator Terminal Bushings

Condenser bushing or solid duct type terminal shall be provided to terminate both ends of each phase winding outside the stator housing. The bushing mounting flanges shall be non-magnetic.

The joints between the bushings and the isolated phase bus shall be bolted. A neutral star-point bar shall also be provided in accordance with the above requirements.

Where applicable the bushings shall be sealed against hydrogen leakage and designed for a pressure inside the generator of 8 kg/cm² or 1.5 times the maximum operating pressure, whichever is greater.

The phase and neutral bushings or terminal shall be arranged in two (2) transverse rows and the connection shall be identified U₁-V₁-W₁ respectively.

4.11 Bushing Current Transformers

The following current transformers shall be provided on the generator terminals for protection relaying and for metering. Three (3) current transformers on each neutral side terminal with five (5) or one (1) amp secondaries.

Two (2) current transformers on each line terminal with five (5) amp or one (1) amp secondaries.

The relay circuit current transformers shall be class 5P20 accuracy class, or better, and the metering current transformers class 0.2, or better. Current transformers for metering shall be calibrated in the factory. Calibration curves shall be furnished.

4.12 Rotor

The rotor including field poles, shaft and couplings shall be machined from single solid steel forging which is homogeneous throughout. Alternatively, the turbine end coupling may be manufactured separately and shrunk onto the rotor shaft.

The Contractor shall provide any special equipment or facilities required to withdraw the generator rotor. Couplings between turbine and generator rotors and all other couplings shall be completely enclosed and covered by guards to give protection to operators.

The rotor shall be grounded.

4.13 Rotor Winding

Generator rotor windings shall be insulated with mica or other suitable material specially chosen to ensure that deterioration by temperature, vibration, or other causes due to constant running under all working conditions is kept to a minimum. All insulation and packing materials shall be non-hygroscopic.

The rotor winding conductors shall be of silver bearing copper.

All cross connections between field coils and connections between coils and collector rings, if used, shall be of ample cross section, carefully insulated, rigidly supported and fastened so as to prevent undue vibration, wear and breakage. All live parts shall be insulated. The windings, including the overhangs, shall be braced and wedged to withstand all specified operating conditions including over speed.

The ground insulation shall also be adequately supported in all positions.

All live parts connected to the field winding, including joints, links, terminals, connections inside the shaft, shall withstand the same high-voltage tests as the field winding. In addition, they shall be fully rated for generator specified operating conditions, and able to withstand centrifugal forces and differential expansion without failure or creep.

The main winding slot wedges shall be non-magnetic, and shall not yield or deform when subjected to the rotor field winding or damper temperatures that occur under specified operating conditions.

Bidder shall state the damping torque of the complete rotor, to a tolerance and a qualification which he shall give, for system analysis purposes. The damping torque shall be defined as the synchronous motor torque (per unit of rated load generator torque) at 0.02 per unit slip.

The rotor end winding retaining rings shall be of non-magnetic austenitic work hardened steel. The retaining rings shall be designed and manufactured to withstand the forces due to fitting to the rotor body, and forces imposed by the end windings and centrifugal stress during operation. The retaining rings shall be securely attached to the rotor body and there shall be no movement relative to the shaft at normal speed.

By using the Contractor's Instruction Manual and his recommended Spare Parts, it shall be possible to carry out the following operations on site in order to repair or inspect the winding.

- (1) Removal and replacement of end winding retaining rings.
- (2) Removal and replacement of the insulation cylinder between the end winding and the retaining ring.
- (3) Lifting out of any coil and its replacement without damaging the slot insulation, and preferably without damaging end winding insulation.

The method of removing and replacing the retaining rings shall be given. All necessary tools and parts for this operation shall be supplied, and the ring heating methods and heating capacity to be supplied shall be stated.

4.14 Generator Field

Brush gears shall be provided for the main field connections in the case of a static excitation system (if proposed) and for rotor field ground detection equipment or other instrumentation as applicable in the case of a brushless excitation system. The following requirements apply mainly to equipment required for the main field connections of a static excitation system.

The slip rings shall preferably be helically grooved and shall be turned on the final shaft center following shrinking on.

The brush gear shall be mounted in an enclosure separate from the main generator casing and not subject to ingress of bearing or oil seal vapor. The brush gear shall be air-cooled, self-ventilated and filtered such that specified brush gear temperatures are not exceeded and servicing is minimized.

The brushes shall be sized for continuous field currents corresponding to generator overload with one (1) brush holder not in service. The connection of brush lead or pigtail to the brush itself shall be of a proven type, and shall not set up arcing.

Slip ring brushes shall be of such number, design and spacing that they can be removed one (1) at a time, maintained or replaced while the generator is running under load.

4.15 Rotor Fans

Shaft mounted fans of the axial type shall have removable blades, preferably of the variable pitch type. Shrouds shall preferably be radially split.

4.16 Bearings

All generator bearings shall be of the journal type, with horizontally split shells for inspection of white metal, enabling replacement of the bearing halves with minimum shutdown time.

Details of the process of bonding and keying the white metal to the shells shall be submitted to the Owner for quality assurance inspection purposes. Every bearing shall be separately identified, even where interchangeability is guaranteed.

Every bearing, including those of exciters, shall be insulated against the effect of shaft voltages and currents that may occur due to normal magnetic unbalance, or due to specified generator electrical faults and resulting added magnetic unbalance.

Bearings shall be suitably insulated where necessary to prevent the flow of shaft current.

4.17 Generator Excitation System

The generator excitation system shall be either brushless or static rectifier type using an external excitation transformer, based on his proven design and service experience for generators of comparable size and ratings.

The excitation system voltage response ratio shall be at least 2.0. The Bidder shall furnish all data pertaining to the generator and its excitation system which depend on the voltage response ratio separately for the specified response ratios of 2.0

The system shall provide for automatic and manual control of the generator terminal voltage, and shall include all the specified or required components, protection, controls and instrumentation.

This Specification covers the general requirements of both types of the excitation system; some parts are common to both the systems and others apply to only one (1). The Bidder shall therefore interpret the excitation system requirements in this specification accordingly.

The generator will be connected via a generator transformer into 500 kV systems to which other generation equipment is connected and which is also interconnected with a similar system operating at 500 kV. System switching will include single- pole fast auto reclosing and three-phase delayed auto reclosing. This shall be taken into account in the design of the excitation system.

All accessory equipment that is required to make a complete excitation system shall be furnished including ground detection equipment, bus connections, and cooling system. Each thyristor shall have the fuse for disconnecting at failure condition. Ground detection equipment shall be provided in excitation cubicle or protection relay cubicle.

The excitation system and the generator shall be capable of operating at ceiling excitation (with voltage of 140% of the field voltage required to

operate at rated kVA) for 30 second according to IEC, without injurious heating.

A fused field flashing circuit shall be provided. The Contractor shall provide the necessary field flashing equipment to build the generator field on start-up.

4.17.1 Voltage Regulating Equipment

The equipment as specified below shall be furnished and shall be so designed that the failure of the regulator to function properly will automatically sound an alarm and disconnect the regulator from service. The generator shall, however, continue in operation under manual control. The manual control range shall permit setting of the generator terminal voltage between a minimum of 50% of nominal volts on open circuit and a maximum of 105% of nominal volts at full load.

The automatic voltage regulator (A.V.R.) shall be capable of performing the following functions:

- (1) Voltage control accuracy $\pm 0.5\%$ on a single running unit with negligible dead band.
- (2) Controlled voltage setting range $\pm 10\%$ adjustment at full load
- (3) Volts per Hertz limiting.
- (4) Transformer reactance drop compensation adjustment range $\pm 5\%$
- (5) Adjustment of phase and feed back elements.
- (6) Maximum excitation limiting.
- (7) Automatic changeover to manual control on initiation from protective devices.
- (8) Spare terminals to allow connection of supplementary control and stabilizing circuits.

The Contractor shall provide calculations to demonstrate the ability of his excitation system to reach and sustain ceiling voltage for bolted three-phase fault on the 500 kV bus at Muara Tawar.

The A.V.R. shall sense the averaged three-phase generator terminal voltage and shall control the exciter output voltage such that a two per cent (2%) deviation in terminal voltage above or below a set point will cause the exciter to force to its positive or negative ceiling voltage.

The thyristor firing control circuits shall be capable of normal operation down to 35% of rated generator terminal voltage to provide proper control during system faults typically of 15-20 cycles duration.

The A.V.R. shall have an adjustable automatic minimum field current control to maintain stability of the generator under all load conditions and to prevent exceeding thermal limitations at leading power factors. Minimum excitation shall be controlled without variation due to field temperature. The A.V.R. shall also have adjustable, inverse time, means to detect that thermal time limitations of the generator are being approached, while overexcited. Control equipment shall be provided to place the generator excitation on manual control after the generator has operated at or near ceiling excitation for a period of time approaching the thermal inverse time limitation of the generator.

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A voltage tracking panel shall be included in the regulator circuit to automatically adjust the manual regulator to keep its output the same as the automatic regulator.

The Contractor shall supply necessary appurtenances connected to voltage balance relays so that the voltage regulator will be tripped from automatic to manual upon operation of balance relays. Balance relays shall be connected between two (2) sets of generator voltage transformers to detect voltage transformer circuit failure. In addition the automatic voltage regulator shall have equipment to detect and trip from automatic to manual regulation in the event of prolonged over excitation.

The voltage regulator shall also include:

- (1) A mechanically latched device, electrically closed and tripped on 220 V DC, to transfer voltage regulator system from automatic to manual control.
- (2) Crosscurrent compensators.

- (3) Necessary stabilizers.
- (4) Fused disconnect switches for AC and DC control circuits.

The regulator shall have on-line monitor and test panel and also provision for static switching between redundant power supplies.

A power system stabilizer (PSS) shall be provided, complete with the transducers necessary to provide the input signals for supplementary control and stabilizing signals to the excitation system. The PSS output characteristics, including frequency sensitivity range, band width and filter adjustment details shall be given in the Bid.

4.17.2 Static Excitation System

If the generator uses the static rectifier type excitation system, this equipment shall also include the followings:

The static exciter rectifier shall consist of thyristor bridges connected in parallel and rated such that with one of the parallel bridges out of service, the remainders are capable of supplying full load excitation requirements with full forcing capability.

Thyristors shall be protected by fuses and surge limiting circuits and facilities shall be furnished for isolating and replacing each thyristor without reducing load on the generator. A local indication system and contacts for remote alarm shall be provided for detecting blown thyristor fuses. Spare modules shall be supplied.

The exciter field rectifier (if furnished) shall be provided with overcurrent relays. Each power rectifier section shall be equipped with neon light indication of blown fuse, disconnecting switches and seal in temperature alarms. The rectifiers shall also be monitored for proper conduction and any malfunction shall be alarmed.

Local panel mounted ammeters shall be provided to indicate current in each bridge.

4.17.3 Excitation Transformer (if Required/Proposed)

The excitation transformer shall be indoor dry type and shall conform to IEC 60076. The excitation transformer shall be fed from the isolated phase bus duct.

The transformers shall be supplied with one (1) current transformer on each of the high-voltage bushings. The current transformers shall comply with the requirements of IEC 60044.

4.17.4 Field Circuit Breaker

Contractor shall furnish enclosed exciter field switch and exciter control panel. The following equipment shall be mounted in the control panel:

- (1) One (1) adequately rated field circuit breaker of AC air or vacuum circuit breaker, drawout type, suitable for 220 V DC control, each complete with field discharge resistor contacts to close before the main contacts open, shunt trip coil, control relay, and minimum of eight (8) auxiliary switches.
- (2) Emergency operating handle for circuit breakers.
- (3) Fused disconnecting switches for DC control circuits.
- (4) Minimum of three (3) sets of potential fuses for instruments, one (1) for exciter voltage, one (1) for generator field voltage, and one (1) for ground detector.
- (5) Generator field ground detector relays shall be suitable for alarm or trip circuits.
- (6) Mounting and main copper connections for: Two (2) shunts, one (1) for generator/exciter alternator field current transducer and one (1) for use with over excitation relays.
- (7) Shunt for main generator/exciter alternator field current in the positive lead with fused shunt outputs.
- (8) Bypass contacts required to complete the interlock circuits when breakers are in drawout positions.
- (9) Necessary clamp type terminal lugs, cable supports and main copper connections.
- (10) Terminal blocks, nameplates and all internal wiring.

Coincident with a field breaker trip signal, the thyristor firing controls shall be phased back to minimum to reduce the duty on the field circuit breaker contacts.

The field circuit breaker shall be supplied for local and remote electrical close and trip. A field ground detector with test circuit shall be provided and fitted in the field circuit breaker cubicle. Indicating lamps for open-close indication of the field circuit breaker shall be mounted on the front of the cubicle.

4.17.5 Generator Excitation Panel

The voltage regulator equipment and the exciter field breaker thyristors, etc. shall be furnished mounted in a group of free standing, sheet metal panels.

All main field connections between the excitation transformer, thyristor rectifiers, field circuit breaker and generator brush gear shall be provided in the form of non-segregated bus duct.

4.17.6 Brushless Exciter and Rotating Rectifier

If the generator uses the brushless exciter, it shall meet the requirements of the applicable parts of IEC 60034, unless otherwise specified herein.

The exciter and rotating rectifiers shall be fully accessible for maintenance. The exciter shall be air cooled, with built-in water cooled heat exchangers arranged such that two (2) separate cooling paths are provided for the exciter itself and the rotating rectifiers. Bearings shall be insulated against shaft currents.

The exciter windings shall be insulated with Class F although temperature rises and total temperatures shall not exceed those permitted for Class B.

Diodes shall be of the silicon type, mounted in compression against centrifugal force.

In the event of failure of one (1) diode, the generating set shall be able to continue running safely at guaranteed generator ratings until the next scheduled servicing shutdown. Diode failure detection equipment shall be

provided.

Means shall be provided for the measurement and indication of exciter field current and voltage, main generator field current, voltage, and temperature. Main generator field ground fault detection equipment shall also be provided.

4.17.7 Excitation System Protection and Control

The field rheostat and voltage adjusting rheostats shall be motorized for DC operated remote control. Suitable outputs shall be included for remote field voltage and current and A.V.R. balance indicators also, on the LCD Monitor located in Central Control Room.

Alarm and trip contacts, where appropriate, shall be provided to indicate at least the following conditions:

- (1) Field breaker open
- (2) Field breaker closed
- (3) Field ground fault
- (4) Cooling air failure
- (5) Exciter load being reduced
- (6) Automatic voltage regulator trip
- (7) Rectifier DC overcurrent
- (8) Exciter trip
- (9) Loss of trip circuit continuity
- (10) Loss of DC control voltage
- (11) Failure to energize field flashing circuit
- (12) Fuse failure
- (13) Field discharge initiated
- (14) Loss of AC and/or DC power supply
- (15) Mode of operation-auto or manual
- (16) Under excitation limit
- (17) Air flow differential high
- (18) Over-excitation limit and trip

- (19) PSS fault
- (20) V/Hz limit
- (21) Auto follower fault
- (22) Loss of firing pulse
- (23) A phase fault detecting system shall also be provided for the exciter (if applicable).

If any contacts external to the turbine generator systems are required for interlocks, the Contractor shall provide them.

The A.V.R. shall include a protective device with two (2) alarms to prevent the regulator going to ceiling voltage on loss of generator terminal voltage signal in the event of generator potential transformer fuse failure. The A.V.R. shall transfer to manual operation in case of loss of voltage signal or loss of automatic voltage regulation function.

Excitation system output overcurrent protection and limiting shall be supplied to provide tripping contact outputs in the case of overcurrent caused by a fault in the control system.

DC voltage and current transducers shall be supplied to provide excitation level output for indication, protection and alarm purposes.

4.17.8 Operation

Under normal conditions the generator shall be started and stopped with the excitation control system switched to the automatic mode. For the static excitation scheme the circuit design and control equipment shall be such, that at approximately 95% speed on start-up, the field flashing supply is applied to the generator rotor to promote a buildup of terminal voltage. At this point the automatic excitation control system shall assume control and the flashing supply be de-energized. Failure to excite after a predetermined time shall cause the flashing supply to be disconnected and an alarm to be initiated. On generator shut-down, the excitation system shall be phased back to minimum excitation when the generator speed falls below 90%. For the brushless excitation, the control system shall build up the generator voltage automatically at a predetermined speed, and similarly reduce the voltage

during run-down. The Contractor shall provide a complete logic diagram showing the sequence of the operations of the excitation system.

The Contractor shall provide the range and approximate final settings of all parameters. The Contractor shall clearly show any restrictions to his generator capability curves caused by the excitation system.

The Contractor shall perform Stability Study to ensure the suitability of his excitation system to the PLN Network system.

4.18 Generator Auxiliary Systems

4.18.1 Generator Hydrogen Supply and Cooling System (If applicable)

If the generator is cooled by hydrogen system, a complete hydrogen gas supply system shall be provided for each generator. The equipment shall include circulating fans, gas leakage measurement equipment, hydrogen coolers, all piping, valves, control and indicating devices for filling the generator with hydrogen. The system shall also automatically maintain the correct stator gas pressure, purity and humidity levels during operation and shall be suitable for purging the frame using carbon dioxide. The automatic gas dryer shall continuously dry the hydrogen. The dryer shall automatically regeneration of the silica gel.

Hydrogen shall be provided from a central production and storage plant to be supplied under this Contract. The Contractor shall provide connections to the hydrogen supply. These connections shall deliver the hydrogen to the hydrogen control panel, where the pressure shall be reduced and kept constant by a pressure regulator.

In addition to the main hydrogen supply system, a bottle rack shall also be provided adjacent to the hydrogen control panel. The bottles shall provide hydrogen for the generator during commissioning or emergency conditions when the main hydrogen supply system is out of service. The bottle rack shall be designed to accept sufficient bottles to provide the generator hydrogen make-up requirement for at least one (1) day. An adjacent storage rack with spare bottles shall also be provided. The total bottles provided for the unit shall have a capacity equal to the normal leakage from the unit for a period

of three (3) days. The hydrogen bottles shall be of standard portable size and shall be provided by the Contractor.

A complete system with all accessories and control devices shall be furnished for automatic control of the hydrogen gas for generator cooling. This shall also include equipment for filling, emptying and purging the generator housing and for automatically purifying and drying the hydrogen to maintain the required density within the generator and cooling system.

The generator hydrogen cooling system shall be of the closed loop, hydrogen circulation type. The hydrogen shall be cooled by the closed cooling water system through hydrogen coolers mounted in the stator casing.

The generator hydrogen cooler shall contain the necessary number of section and be ample capacity to serve the enerator. Coolers shall be sized to allow operation at not less than 67% of generator kVA rating with one (1) section out of service.

The Contractor shall supply and install the necessary 100 ohm platinum resistance temperature detector(s) for control purposes.

Six (6) thermocouples complete with wells for generator gas temperature measurement shall be furnished. Temperatures shall be monitored from the Hydrogen Control Panel specified herein.

4.18.2 Generator CO₂ Supply System (If applicable)

A carbon dioxide supply manifold shall also be furnished for the complete purge of the generator.

Devices shall be provided for reducing and automatically regulating the supply for distribution to nozzles in the bottom of the generator stator casing. The carbon dioxide will be stored as a liquid, but shall be distributed in a gaseous state, and the system shall be designed to prevent restriction of the gas flow to the generator.

The Contractor shall be responsible for the supply of carbon dioxide for the generators until the end of the warranty period.

4.18.3 Hydrogen Control Panel

One (1) hydrogen control panel and associated instrumentation for controlling and continuously monitoring the hydrogen coolant shall be furnished for each generator. Each panel shall be furnished complete with all the necessary regulating and controlling devices, indicators, alarms, annunciator, continuously rated alarm horn, purity, temperature & pressure gauges and recorders, etc.

The generator hydrogen purity shall be maintained continuously at not less than 98% without exceeding the guaranteed hydrogen consumption.

The following local and remote alarm shall be provided, among others, at the hydrogen control panel:

- (1) Presence of moisture/water in the generator.
- (2) High temperature of hydrogen leaving the cooler (only in control room)
- (3) Low and high pressure of hydrogen supply.

4.18.4 Seal Oil System (If applicable)

The seal oil system shall include all pumps, motors, coolers, detrainig tanks, piping, valves and float traps. The system shall be arranged to automatically maintain the oil at the shaft seals at a pre-set positive pressure above that of the hydrogen in the stator casing. Oil which drains from the hydrogen side seals shall be collected in detrainig tanks. Vacuum treating equipment shall be provided to completely remove entrained gases and water from the seal oil.

The generator shall have shaft seals at both ends to prevent hydrogen leaking from the stator casing through the circumferential clearances between the casing end shields and the shaft. The shaft seals shall be designed to accommodate full axial expansion and construction of the generator shaft and shall be properly keyed to prevent twist and rotation. All hydrogen seal oil equipment shall be of steel construction.

Thermocouples or RTD's shall be installed in the oil inlet for continuous monitoring of seal oil temperature.

The normal seal oil pump shall be either turbine generator shaft driven or an AC motor driven. The Contractor shall design, supply and install the necessary control system for these pumps. Cleanable oil filters shall be installed in the oil supply pipe line close to each seal.

The Contractor shall supply and install the necessary control system and valve(s) to automatically control the flow of cooling water to maintain the oil at the temperature required by the Contractor.

The annunciator shall be furnished with contacts wired to signals “Hydrogen Supply and Cooling System” troubles to the plant computer alarm system.

4.18.5 Generator Closed Air Cooling System (If applicable)

If generator is cooled with air cooling system, the generator cooling system shall include all necessary coolers, fans, pumps and control apparatus as appropriate and comply with the requirements of IEC 60034.

The system classification shall be complete with all necessary equipment for safe and efficient operation of the plant at normal working pressures.

The generator shall have a closed air circuit cooling system with stator and rotor cooling requirements as specified in the section for Stator and Rotor. Heat shall be extracted from the closed circuit by at least two (2) air/demineralized water heat exchangers.

The heat exchanger shall comply with the relevant requirements of the heat exchanger specified under Auxiliary Cooling Water System.

Suitable temperature indication, high temperature alarm and trip instrumentation shall be provided as specified for Hydrogen Cooling System.

4.19 Generator Auxiliary Equipments

4.19.1 Voltage Transformer and Surge Protection

Voltage Transformer

(1) Type : Dry/Encapsulated

- (2) Rated voltage : Generator voltage / 110 V
- (3) Operating voltage range/
ratio (Max/Normal) : 1.05
- (4) Frequency : 50 Hz
- (5) Accuracy class : 0.2 meeting the requirements for Class
3P at low voltages
- (6) Thermal burden at system
line to line voltage each
secondary winding : 500 VA/phase (1500 VA total)
- (7) Insulation Level
 - a. Rated impulse
Withstand voltage : as per generator voltage *)
 - b. Rated power-frequency
short duration withstand
voltage (r.m.s.) : as per generator voltage *)

Note: *) Items to be determined by the Contractor and subjected to the
Owner's Approval.

Surge Protection

Type : Gapless (Zinc-Oxide)

Note:

Bidder shall submit full calculations for selection and sizing of the surge
protection equipment taking into account the possibly of surge transfer from
the Network, for Owner's review.

At the generator side, voltage transformers shall be connected in two (2)
groups of three (3) transformers in each group. At the generator transformer
side, voltage transformers shall be connected in one (1) group of three (3)
transformers.

Voltage transformers shall be designed for operation on a nominal generator
voltage with generator neutral grounded through high impedance.

Each set of voltage transformer shall be assembled with its protective fuses in a separate compartment. In the withdrawn position, voltage transformers and fuses shall be completely disconnected from service and all exposed parts visibly grounded. Fuses shall be furnished on both primary and secondary sides of the voltage transformers.

One (1) set of surge protection equipment per phase shall be mounted in the voltage transformer cubicle and shall be connected between each phase and ground for the protection of the generator circuit. The surge protection equipment shall consist of surge arresters and capacitors, bushings and all other necessary accessories.

The primary circuit of the surge protection equipment shall be branched from the voltage transformers. Each surge arrester shall be fitted with surge counter.

4.19.2 Generator Neutral Grounding

4.19.2.1 Grounding Transformer

- | | | |
|--|---|--------------------------------|
| (1) Type | : | Dry/Encapsulated, self-cooled. |
| (2) Rated Voltage | : | Generator voltage / 240 V |
| (3) Frequency | : | 50 Hz |
| (4) Rated time | : | 5 minutes |
| (5) Temperature rise
above 40°C ambient | : | 150°C |
| (6) Insulation class | : | H |
| (7) Insulation level | | |
| a. Rated impulse withstand
voltage (peak) | : | as per generator voltage *) |
| b. Rated power-frequency
short duration withstand
voltage (r.m.s.) | : | as per generator voltage *) |
| (8) Service | : | Indoor |

Note: *) Items to be determined by the Contractor and subjected to the Owner's Approval.

4.19.2.2 Grounding Resistor

- | | | |
|--|---|------------------------------|
| (1) Type | : | Cast iron grid resistor |
| (2) Mechanical current | : | 1.2 x thermal current rating |
| (3) Rated voltage | : | 240 V |
| (4) Frequency | : | 50 Hz |
| (5) Rated time | : | 1 minute |
| (6) Temperature rise
above 40°C ambient | : | 500°C |
| (7) Service | : | Indoor |

Note: The Contractor shall submit full calculations for sizing the grounding transformer, grounding resistor for Owner's review.

The grounding resistor shall be connected across the secondary of the transformer. An insulating barrier plate shall be provided on the roof of the generator neutral cubicle together with bus connectors to provide adequate support for the neutral bus.

The generator neutral grounding cubicle shall be ventilated and shall be constructed of non-magnetic material.

Two (2) heavy duty thermostats or equivalent devices shall be furnished in each generator neutral cubicle to monitor the heat generated by the neutral resistor due to sustained current flow resulting from an undetected or uncleared stator a ground fault.

4.19.3 Auxiliary Equipment Cubicles

The cubicles specified for housing voltage transformers, surge protection equipment, generator neutral grounding, etc., shall be of isolated phase type construction.

The cubicles shall be furnished with hinged doors or roll-out drawers, for each phase, to permit access to equipment for inspection and repairs. The Contractor shall provide and install each cubicle with channel steel bases and mounting bolts.

The cubicles shall be designed to withstand the gas pressure produced by internal bus faults.

All wiring within the cubicles shall be provided and installed by the Contractor with terminations for control cable connections made on terminal blocks located near the top of the cubicles in a suitable terminal cabinet unless otherwise specified.

The nameplate and caution plate shall be attached at a position where it can be seen easily from the floor level. Thermostatically controlled electric anti-condensation space heaters shall be provided inside each cubicle. Each heater shall be controlled from a double-pole switch with white lamp labeled "Cubicle Heater On".

4.20 Miscellaneous Electrical Requirements

Starters for the 3-phase AC motors shall be furnished in motor control centers except where specified otherwise. All electrical equipment and wiring inside the Contractor's equipment shall meet the requirements of the Technical Specifications E-001.

5 PERFORMANCE TESTING AND GUARANTEE

5.1 Calculations and Type Tests

5.1.1 Generator Calculations

The Contractor shall describe the calculation methods and results for the following items:

- (1) Hydrogen consumption (if applicable)
- (2) Generator reactances and time constants
- (3) Efficiencies (the method and tolerances shall be subject to the approval of the Owner).
- (4) Rated load excitation

- (5) Calculated temperatures at rated load.

5.1.2 Generator Type Tests

Generator type tests shall be performed. The Contractor may submit generator type test data based on tests on machines similar to those proposed for the Owner's review and acceptance. If type test results for similar machines are rejected, the Contractor shall perform the necessary type tests on the generator at his expense. During the type tests, temperatures, coolant flows and lubricant flows shall be as near as possible to normal operating conditions. Where this is not possible, the effect of normal operating conditions shall be explained and steps taken to eliminate their effect. Type tests are required for the following items:

- (1) Unexcited heat run
- (2) Generator heat run at rated voltage with the machine open circuited.
- (3) Generator heat run at rated current with the generator short circuited.
- (4) Sudden short circuit test to determine the generator reactances and time constants. The short circuit test shall be carried out starting from a no load condition with 50% terminal voltage.
- (5) Measurement of zero sequence reactance
- (6) Measurement of negative sequence reactance
- (7) Excitation System

5.2 Shop Tests

The generator components shall be shop tested and the generator shall be shop assembled to the degree required to ensure proper operating clearances, proper fit up of components for field erection and proper overall operation.

5.2.1 Overspeed Tests of Rotors

Each complete rotor shall be subjected to an overspeed test of 20% above rated speed for 2 minutes as per IEC 60034. Any sign of damage or distortion shall be a cause for rejection. Each rotor shall be statically and dynamically balanced before and after overspeed testing.

5.2.2 Generator and Auxiliaries

Before assembling the stator core, the housing shall be subjected to a water pressure test of 8 kg/cm²(g) or 1.5 times the maximum operating hydrogen pressure, whichever is greater for 15 minutes. If the casing is one (1) piece, it shall be tested with the end shields bolted in position as for normal operation. A factory core loss test or ring test shall be done for each core to measure the core iron loss and to demonstrate absence of hot spots, after final compression of the complete core assembly.

The bushings shall be tested separately from the test of the generator. Testing shall be done in accordance with IEC 60137.

The generator shall be assembled at the Contractor's factory for testing. Testing shall be carried out in accordance with ANSI/IEEE 115. During testing, the temperature, pressure, coolant flow and lubricant flow shall be as close to normal operating conditions as possible. The following tests shall be performed:

- (1) Hydrogen gas leakage measurement (if applicable).
- (2) Bearing insulation resistance measurement.
- (3) Shaft current measurement.
- (4) Bearing vibration measurement.
- (5) Generated wave form and harmonic analysis.
- (6) Telephone influence factor measurement.
- (7) Voltage balance.
- (8) Phase sequence.
- (9) High voltage dielectric tests on stator and rotor in accordance with ANSI/IEEE C50.10 and ANSI/IEEE 95, or equivalent.
- (10) Winding resistance measurements on stator and rotor.
- (11) Insulation resistance measurement on stator and rotor (Refer to Sections 4.9 and 4.13).
- (12) Capacitance and loss tangent measurement of stator winding.
- (13) Measurement of open circuit characteristics and losses.
- (14) Short circuit characteristics and losses.

- (15) Measurement short circuit ratio.
- (16) Measurement of direct axis synchronous reactance.

Rotating exciters shall receive the following factory tests:

- (1) Winding resistance measurement.
- (2) Open circuit characteristics and losses at voltages up to ceiling voltage.
- (3) Short circuit characteristics and losses.
- (4) Core loss heat run.
- (5) Load loss heat run.
- (6) Mechanical loss heat run.
- (7) Exciter response ratio measurement.
- (8) Vibration measurements.
- (9) Dielectric tests in accordance with IEEE 95.
- (10) Functional check of the operation of the voltage regulator under both manual and automatic operation.
- (11) Functional check of all voltage regulator control and protection systems.
- (12) High-voltage dielectric tests of all voltage regulator wiring in accordance with dielectric tests mentioned in the paragraph below.

Static excitation system components shall be subjected to the following tests.

- (1) Transformer heat run and loss measurement.
- (2) Thyristor rectifier heat run and loss measurement.
- (3) High-voltage dielectric tests.
- (4) Field circuit breaker operation tests.
- (5) Measurement of the excitation system response ratio and ceiling voltage.

Dielectric tests shall be carried out according to the following:

- (1) High-voltage tests on control and auxiliary wiring including all temperature detectors and wave form coils shall be 1500 V.
- (2) The acceptance proof-tests, shall be per ANSI C50.10 and IEEE 95.
- (3) All high voltage tests shall be made between the winding under test and the machine frame, which shall be solidly grounded and to which all other windings and wiring shall be connected during the test.

5.3 Site Tests

The main generator windings shall withstand 85% ANSI factory acceptance proof test voltage for one (1) minute after the machine has been brought up to synchronous speed and allowed to come to rest. Following the above test, the unit shall be run up to synchronous speed and overexcited to deliver 105% of generator rated terminal voltage for 30 seconds just prior to synchronizing with the Owner's system, with the neutral of the generator grounded through the neutral grounding transformer and resistor.

Tests of all other equipment which are further defined in Section 4, shall be in accordance with IEC standards or equivalent unless otherwise specified. The following tests shall be performed:

- (1) Dry out of the generator.
- (2) Test of the generator open and short circuit characteristics.
- (3) Generator vibration measurements.
- (4) Generator phase sequence and terminal voltage balance.
- (5) Measurement of shaft voltage.
- (6) Line charging capacity.
- (7) Full load temperature rise.
- (8) Measurement of hydrogen gas leakage (if applicable).
- (9) High-voltage tests on excitation system busbars, including metal enclosed bus, if applicable.
- (10) The resistance of all joints in the exciter metal enclosed bus shall be measured, if applicable.
- (11) All high and low voltage circuits shall be insulated resistance tested.

- (12) Current injection or other tests to verify current transformer ratio and polarity.
- (13) Testing of the associated relaying and metering equipment including resisting with the tests on metering and line protection equipment.

The Contractor shall verify the correct performance and correct settings of his excitation equipment by site tests. These tests shall include, but not be limited to, the followings:

- (1) Demonstration of exciter control during load changes.
- (2) Verification of exciter rectifier failure detectors.
- (3) The excitation system response for step error signals shall be verified at no load and selected loadings.
- (4) The exciter voltages shall be measured at both high and low generator terminal voltages.
- (5) The voltage regulator error signals to cause maximum positive and negative field voltages shall be measured.
- (6) The operation of all excitation system protective functions shall be verified.
- (7) The transfer from automatic to manual regulator control shall be verified.
- (8) The Contractor shall perform tests to verify the correct operation of his power system stabilizer.

5.4 Performance in Connection to Turbine

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The Contractor shall conduct performance tests to prove the capacity and efficiencies guaranteed by the Contractor according to the applicable Standards & Codes. The Contractor shall furnish a test procedure for testing the equipment. The test procedure shall clearly spell out the instrumentation, test connections, cycle isolation requirements, and any other equipment for the tests and where they must be located in the system to obtain the necessary test data. The Contractor shall furnish all the required temporary and permanent testing and isolating equipment and instrumentation. The Contractor shall be responsible for contracting an independent party mutually acceptable to the Owner to record the test data, calculate the required

parameters and furnish the Owner with a complete report of the test results. The Contractor shall also arrange to have his authorized representative(s) witness and assist in these tests.

6 QUALITY ASSURANCE/QUALITY CONTROL

6.1 General

Quality assurance/quality control of the generator, its subsystems and appurtenances specified herein this Specification shall conform to the requirements of Part 4, Subsection 2.7.

Quality assurance/quality control procedures so specified shall include, but not be limited to, the following:

- (1) Procurement
- (2) Forging and casting
- (3) Welding
- (4) Heat treating
- (5) Radiography
- (6) Packaging and shipping
- (7) Testing
- (8) Cleaning and flushing
- (9) Non destructive testing

Quality assurance/quality control documentation as specified shall include, but not be limited to, the following:

- (1) Material certificates
- (2) NDE reports
- (3) All shop test reports
- (4) All other documentation required by applicable codes and standards

6.2 Inspection

The Contractor shall provide inspection to ensure the mechanical accuracy of components, compliance with drawings, and identify and acceptability of all material, parts, and equipment supplied under this Contract.

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